A Dynamic Model of Stellar Populations

Modeling Stars With Fixed Lifetimes

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I. INTRODUCTION

Modeling stellar evolution is a crucial aspect of astronomy and astrophysics. Stellar models provide insights into the dynamics of the universe, the process of star formation, and the testing of theoretical models (Aguirre 2018, Bowman et al. 2023, Gallart et al. 2005). Stellar models are also used to inform observational strategies, guiding astronomical surveys. A number of stellar models have been proposed, typically theses models are based on pure theory, or on surveys that can distinguish individual stars, which typically must be local (see, for example, Verro et al. (2022)). There is, however, a gap in population based models. Population based models can be applied to observations of entire galaxies that cannot resolve individual stars. As such, population models could be fit to distant (and therefore old) galaxies, giving insights into stellar evolution over a wider time frame. A particular model for stellar population dynamics was proposed by Bruwel et al. (2024). Bruwel et al. (2024) fit their model to real world data and were able to extract limited estimates for the proportion of a stars mass that returns to the interstellar medium (ISM) at the time of death (p_{ret}), and the rate at which the ISM forms into stars (f).

The model proposed by Bruwel et al. (2024) assumes that stars die according to some rate $r(m) = 1/\tau(m)$, i.e. a constant hazard model. While constant hazard models do have wide applications to real world systems, it is known that stars do not have a constant hazard. Instead, they have a very concentrated spike in hazard around $t = \tau(m)$. In other words, they all die at almost exactly $t = \tau(m)$. The constant hazard does predict the correct mean lifetime, but is inaccurate in all other regards. For example, it predicts that 63% of stars die prematurely before $t = \tau(m)$. This inaccuracy could result in incorrect estimates for p_{ret} and f.

To address this limitation of the former model, this paper introduces a more accurate "fixed lifetime model", where stars die at exactly an age $t = \tau(m)$. This is done by adjusting the model proposed by Bruwel et al. (2024) to be a delay differential equation (DDE). DDEs are a class of mathematical models where the derivatives of the variables at time t can depend on values of the function at a previous time $t - \tau$. This makes them well suited for the problem of fixed lifetimes, and they have been applied to similar problems in biology (see, for example, Feng et al. (2016)).

Beyond proposing a DDE model, this paper simulates from the DDE and compares to the former constant hazard model of Bruwel et al. (2024). Examining if the incorrect specification of the former model could have resulted in incorrect estimates for p_{ret} and f.

II. APPROACHES AND RESULTS

A. Model

The equations presented by Bruwel et al. (2024) are

$$\frac{d\rho(m,t)}{dt} = fM_{\text{ISM}}(t) \cdot \frac{m\Phi(m)}{\langle m \rangle_\Phi} - r(m)\,\rho(m,t) \tag{1}$$

$$\frac{dM_{\text{ISM}}(t)}{dt} = -fM_{\text{ISM}}(t) + p_{\text{ret}} \int_{m_{\min}}^{m_{\max}} r(m)\rho(m,t)\,dm \tag{2}$$

$$\frac{dM_{\text{R}}(t)}{dt} = (1-p_{\text{ret}}) \int_{m_{\min}}^{m_{\max}} r(m)\rho(m,t)\,dm \tag{3}$$

We can rewrite $fM_{\text{ISM}}(t) \cdot \frac{m\Phi(m)}{\langle m \rangle_\Phi}$ as $B(m,t)$ for the birth rate of stars of mass m, and $-r(m)\rho(m,t)$ as $D(m,t)$ for the death rate of stars of mass m. Eqs. (1) – (3) now become

$$\frac{d\rho(m,t)}{dt} = B(m,t) - D(m,t) \tag{4}$$

$$\frac{dM_{\text{ISM}}(t)}{dt} = \int_{m_{\min}}^{m_{\max}} \left[-B(m,t) + p_{\text{ret}} D(m,t)\right] dm \tag{5}$$

$$\frac{dM_{\text{R}}(t)}{dt} = \int_{m_{\min}}^{m_{\max}} \left[(1-p_{\text{ret}}) D(m,t)\right] dm \tag{6}$$

We rearrange Eq. (5) to get

$$\frac{dM_{\text{ISM}}(t)}{dt} = -\int_{m_{\min}}^{m_{\max}} (1-p_{\text{ret}})B(m,t) + p_{\text{ret}}\left(B(m,t) - D(m,t)\right) dm \tag{7}$$

$$= -\int_{m_{\min}}^{m_{\max}} (1-p_{\text{ret}})B(m,t) + p_{\text{ret}} \frac{d\rho(m,t)}{dt} dm \tag{8}$$

Eq. (8) turns out to be more convenient for computational purposes as it does not explicitly involve the death rate.

To modify this differential equation to take into account the fixed lifetime of a star, we simply substitute out $D(m,t)$ for $B(m, t-\tau(m))$ where $\tau(m)$ is the lifetime of a star of mass

m. That is, the number of mass m stars that die at time t is equal to the number of stars that were born at time $t-\tau(m)$. Here we take a brief aside to mention that we could instead analyze $D(t) = -\int_0^\infty B(t-\tau)S'(\tau)d\tau$ where $-S'(t)$ is the probability density function for a star dying at age τ, we are essentially examining the case where $-S'(t) = \delta(t-\tau(m))$, $\delta(t)$ being the Dirac delta function.

Focusing in on Eqs. (4) and (8) (as Eq. (6) is just the remaining mass), we have

$$\frac{d\rho(m,t)}{dt} = F(m)\left(M_{\rm ISM}(t) - M_{\rm ISM}(t-\tau(m))\right) \tag{9}$$

$$\frac{dM_{\rm ISM}(t)}{dt} = -\int_{m_{\rm min}}^{m_{\rm max}} \left[(1-p_{\rm ret})F(m)M_{\rm ISM}(t) + p_{\rm ret}\frac{d\rho(m,t)}{dt}\right] dm \tag{10}$$

Where $F(m) = f \cdot \frac{m\Phi(m)}{\langle m\rangle_\Phi}$ is fixed over time.

B. Computational Approach

To solve Eqs. (9) and (10) discrete bins were used, that is

$$X(t) = \begin{pmatrix} \rho(m_0,t) \\ \rho(m_1,t) \\ ... \\ \rho(m_{-1},t) \\ M_{\rm ISM}(t) \end{pmatrix} \tag{11}$$

Where we take bins that are evenly spaced between $m_{\rm min}$ and $m_{\rm max}$ in log space, and m_0, m_1, ... are the geometric mean of the bin edges. This log scale is used due to the fact that low mass stars live longer, and form more often. Binning was used due to its simplicity, despite its inefficiency. Faster more accurate techniques could be adopted if fitting this model to data.

The integral in Eq. 10 is replaced with a sum yielding the system

$$\frac{d\rho(m_0,t)}{dt} = F(m_0)(M_{\rm ISM}(t) - M_{\rm ISM}(t-\tau(m_0))$$
$$\frac{d\rho(m_1,t)}{dt} = F(m_1)(M_{\rm ISM}(t) - M_{\rm ISM}(t-\tau(m_1))$$
$$\ldots$$
$$\frac{dM_{\rm ISM}}{dt} = -\sum_{m=m_{\rm min}}^{m_{\rm max}} \left[(1-p_{\rm ret})F(m)M_{\rm ISM}(t) + p_{\rm ret}\frac{d\rho(m,t)}{dt}\right]\Delta m$$

Where Δm is the bin spacing.

This system was solved using the integrator developed by Zulko (2023).

C. Simulated Results

Simulations where performed from the system above, and also from the original system from Bruwel et al. (2024). These were done with the same IMF, $m_{\rm min}, m_{\rm max}$ etc. as Bruwel et al. (2024), $f = 0.1Gy^{-1}$, and $p_{\rm ret} = 0.2$. These values were chosen somewhat arbitrarily. The results are shown in Fig 1.

The proposed model follows a similar shape to the original model. However, the amount of mass in stars at any given time is higher. This can be explained by the fact that in the rate model, some stars die prematurely (63%), where this cannot happen with the fixed lifetime.

The similarity in shape indicates that it may be possible to force the former model to align with the new model with a different value for $p_{\rm ret}$ and f.

Fig 2 examines the evolution of stars of a specific mass. It is observed that at the lifetime of the higher mass star there is a sharp change as the stars suddenly begin to die. This would happen to the lower mass star too, except its lifetime is $> 200Gyr$. This sharp change only happens because there is a well defined $t = 0$ where star formation can suddenly happen across the entire galaxy. While this is approximately true for real galaxies, it is not as sudden (see for example Hashimoto et al. (2018)).

III. DISCUSSION

The proposed model is a more realistic description of what happens to stars. However, it still suffers from many of the limitations outlined by Bruwel et al. (2024). Namely

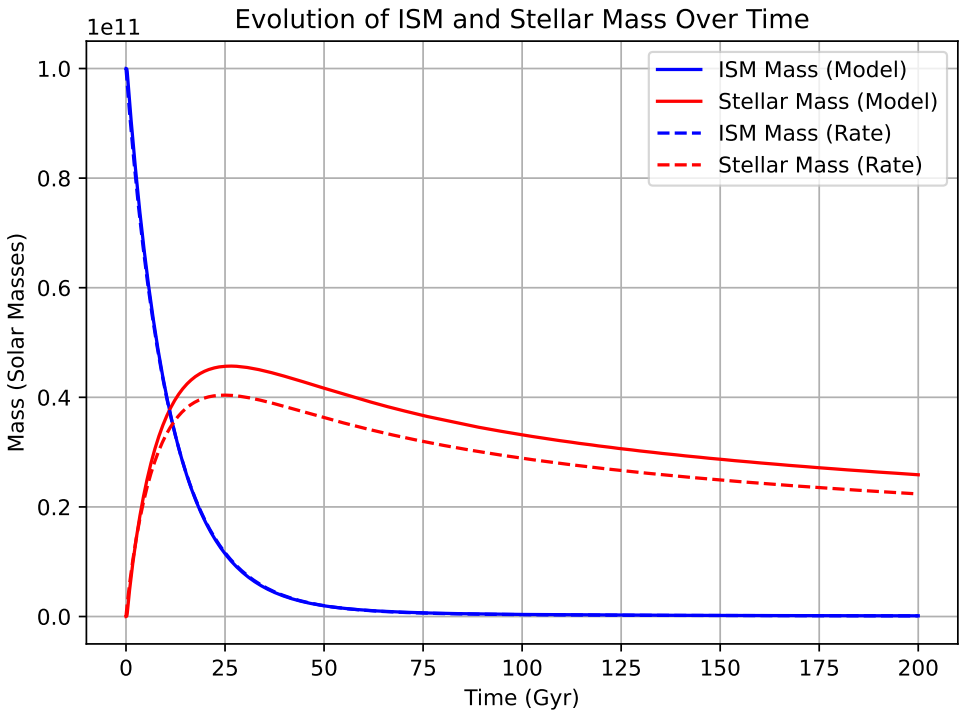


FIG. 1. Comparison of proposed fixed lifetime model (solid) vs former death rate model (dashed).

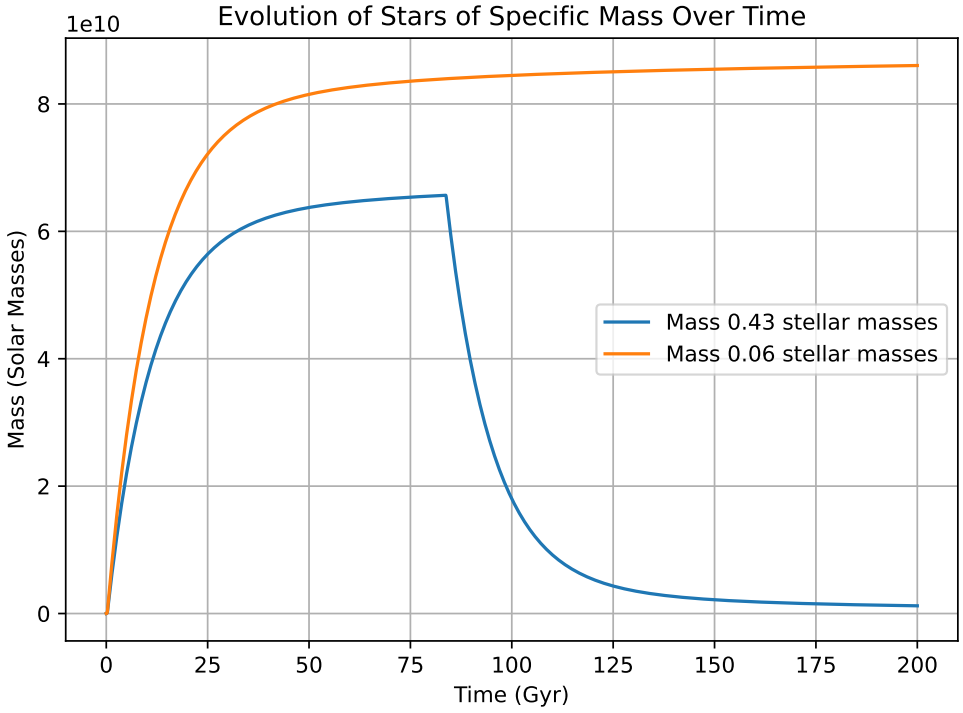


FIG. 2. Total mass in stars of specific mass over time.

1. The only factors of a galaxy are stars, the ISM, and stellar remnants.

2. The proportion of stars that return to the ISM is independent of mass.

3. Stars and stellar remnants do not accrete mass from the ISM.

Limitation 1 can be addressed simply by defining the stellar remnants as being all mass that cannot form into stars. Limitation 2 is a significant limitation, and investigation needs to be done. There is limited existing research on what the functional form of $p_{\text{ret}}(m)$. With sufficient data, the proposed model could be used to determine the form of $p_{\text{ret}}(m)$. Limitation 3 could be easily addressed by adding additional terms to the model. These terms are well understood in astrophysics, see for example Hsu et al. (2022).

IV. CONCLUSIONS

The proposed model provides a better approximation for the evolution of stellar populations by incorporating fixed lifetimes for stars as opposed to a constant death rate. This model demonstrates that by assuming stars cannot die prematurely, there is a higher net mass in stars compared to the former model, and that the shape of the populations over time remains similar. This indicates that the existing model will likely produce incorrect estimates for p_{ret} and f despite potentially having good fit. Despite its improvements, the proposed model still has significant limitations. Future work should focus on refining the model to include mass dependent p_{ret} and accretion.

V. CODE

Code can be found here

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